

Joins and set operations

The screenshot shows the SQL Enterprise Manager interface. The left pane displays the 'joins' database schema, including tables 'class' and 'class_info'. The main pane shows a script with the following SQL statements:

```
1 • create database joins;
2 • use joins;
3 • CREATE TABLE class (
4     id INT,
5     name VARCHAR(30)
6 );
7
8 • CREATE TABLE class_info (
9     id INT,
10    address VARCHAR(30)
11 );
12
13 • INSERT INTO class VALUES
14 (1, 'abhi'),
15 (2, 'adam'),
16 (4, 'alex');
17
18 • INSERT INTO class_info VALUES
19 (1, 'DELHI'),
20 (2, 'MUMBAI'),
21 (3, 'CHENNAI');
22
```

The bottom pane shows the 'Output' window with the following results:

#	Time	Action	Message
81	22:32:08	INSERT INTO Donation_History (donation_id, bo...	1 row(s) affected
82	22:32:08	COMMIT	0 row(s) affected
83	22:32:09	CREATE INDEX idx_books_isbn ON Books(isbn)	0 row(s) affected Re...
84	22:32:10	SELECT * FROM Books WHERE isbn = '978045...	0 row(s) returned
85	22:47:50	use joins	0 row(s) affected

Rohita_2520090235 x

Edit View Query Database Server Tools Scripting Help

SQL

ator: joins x

er objects

bank
book_dbs
bookflow_db
bookflow_dbs
joins
Tables
class
class_info
student_marks1
Views
Stored Procedures
Functions
kleb
klef
istration Schemas
nation

```
17  
18 • INSERT INTO class_info VALUES  
19 (1, 'DELHI'),  
20 (2, 'MUMBAI'),  
21 (3, 'CHENNAI');  
22  
23 • SELECT *  
24 FROM class  
25 CROSS JOIN class_info;  
26
```

Result Grid Filter Rows:

	id	name	id	address
▶	1	abhi	3	CHENNAI
	1	abhi	2	MUMBAI
	1	abhi	1	DELHI
	1	abhi	3	chennai
	1	abhi	2	mumbai
	1	abhi	1	delhi
	1	abhi	3	kolkata
	1	abhi	2	Mumbai
	1	abhi	1	delhi
	2	adam	3	CHENNAI

Result 1

object selected

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SQL Server Enterprise Edition (64-bit) - Microsoft SQL Server 2008 R2

Server Explorer

AS

Server Objects

Server

Database

Tables

class

class_info

student_marks1

Views

Stored Procedures

Functions

Scripts

joins*

```
47
48 • SELECT *
49 FROM class
50 INNER JOIN class_info
51 ON class.id = class_info.id
52
53 • SELECT class.name,
54         class_info.address
55 FROM class
56 INNER JOIN class_info
```

Result Grid

	id	name	id	address
▶	1	abhi	1	DELHI
	1	abhi	1	DELHI
	1	abhi	1	DELHI
	1	abhi	1	delhi
	1	abhi	1	delhi
	2	adam	2	MUMBAI
	2	adam	2	MUMBAI
	2	adam	2	MUMBAI
	2	adam	2	mumbai
	2	adam	2	Mumbai

Result 2 x

object selected

Rohita_2520090235 x

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r objects

bank

book_dbs

bookflow_db

bookflow_dbs

joins

Tables

- class
- class_info
- student_marks1

Views

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Functions

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joins* x

```
53 • SELECT class.name,  
54      class_info.address  
55 FROM class  
56 INNER JOIN class_info  
57 ON class.id=class_info.id;  
58  
59 • SELECT *  
60 FROM class  
61 NATURAL JOIN class_info;  
62
```

Result Grid

	name	address
▶	abhi	DELHI
	abhi	DELHI
	abhi	DELHI
	abhi	delhi
	abhi	delhi
	adam	MUMBAI
	adam	MUMBAI
	adam	MUMBAI
	adam	mumbai
	adam	Mumbai

object selected

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Tables

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student_marks1

Views

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Functions

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ref

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joins* x

59 • SELECT *

60 FROM class

61 NATURAL JOIN class_info;

62

63 • INSERT INTO class VALUES

64 (5, 'ashish');

65

66 • INSERT INTO class_info VALUES

67 (7, 'NOIDA'),

68 (8, 'DAMTOAT');

Result Grid

Filter Rows:

	id	name	address
▶	1	abhi	DELHI
	1	abhi	DELHI
	1	abhi	DELHI
	1	abhi	delhi
	1	abhi	delhi
	2	adam	MUMBAI
	2	adam	MUMBAI
	2	adam	MUMBAI
	2	adam	mumbai
	2	adam	Mumbai

Result 4

object selected

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Tables
class
class_info
student_marks1
Views
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Functions
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Schemas
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object selected

63 • INSERT INTO class VALUES
64 (5,'ashish');
65
66 • INSERT INTO class_info VALUES
67 (7,'NOIDA'),
68 (8,'PANIPAT');
69
70 • SELECT *
71 FROM class
72 LEFT OUTER JOIN class_info
73 ON class.id=class_info.id;
74
75 • SELECT *
76 FROM class
77 LEFT JOIN class_info
78 ON class.id=class_info.id
79 WHERE class_info.id IS NULL;
80
81 • SELECT *
82 FROM class
83 RIGHT OUTER JOIN class_info

Output
Action Output
Time Action Message
8 12:13:55 INSERT INTO class_info VALUES (1,'DELHI'), (2,'MUMBAI'), (3,'CHENNAI') 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
9 12:14:15 SELECT * FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 50000 65 row(s) returned
10 12:14:35 SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id=class_info.id LI... 65 row(s) returned
11 12:14:59 SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 50000 65 row(s) returned
12 12:15:21 INSERT INTO class VALUES (5,'ashish') 1 row(s) affected
13 12:15:21 INSERT INTO class_info VALUES (7,'NOIDA'), (8,'PANIPAT') 2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0

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er objects

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69

70 • SELECT *

71 FROM class

72 LEFT OUTER JOIN class_info

73 ON class.id=class_info.id;

74

75 • SELECT *

76 FROM class

77 LEFT JOIN class_info

78 ON class.id=class_info.id

<

Result Grid



Filter Rows:

	id	name	id	address
▶	1	abhi	1	DELHI
	1	abhi	1	DELHI
	1	abhi	1	DELHI
	1	abhi	1	delhi
	1	abhi	1	delhi
	2	adam	2	MUMBAI
	2	adam	2	MUMBAI
	2	adam	2	MUMBAI
	2	adam	2	mumbai
	2	adam	2	Mumbai

object selected

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objects

ank

ook_dbs

ookflow_db

ookflow_dbs

oins

Tables

class

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student_marks1

Views

Stored Procedures

Functions

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joins x

```
72 LEFT OUTER JOIN class_info
73 ON class.id=class_info.id;
74
75 • SELECT *
76 FROM class
77 LEFT JOIN class_info
78 ON class.id=class_info.id
79 WHERE class_info.id IS NULL;
80
81 • SELECT *
```

Result Grid

	id	name	id	address
▶	4	alex	NULL	NULL
	4	anu	NULL	NULL
	4	alex	NULL	NULL
	4	anu	NULL	NULL
	4	anu	NULL	NULL
	5	ashish	NULL	NULL

object selected

The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Schemas' pane is expanded, showing a tree view of database objects including 'Tables' (class, class_info, student_marks1), 'Views', 'Stored Procedures', and 'Functions'. The 'joins' folder is selected. The main query window displays a SQL query that performs a RIGHT OUTER JOIN between 'class' and 'class_info' tables. The query is as follows:

```

78  ON class.id=class_info.id
79  WHERE class_info.id IS NULL;
80
81  •  SELECT *
82      FROM class
83      RIGHT OUTER JOIN class_info
84      ON class.id=class_info.id;
85
86  •  SELECT *
87      FROM class

```

Below the query window, the 'Result Grid' is visible, showing the results of the query. The grid has five columns: 'id', 'name', 'id', and 'address'. The results are as follows:

	id	name	id	address
▶	1	abhi	1	delhi
	1	abhi	1	delhi
	1	abhi	1	delhi
	1	abhi	1	delhi
	1	abhi	1	delhi
	2	adam	2	Mumbai
	2	adam	2	Mumbai
	2	adam	2	Mumbai
	2	adam	2	Mumbai
	2	adam	2	Mumbai

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SQL Server Enterprise Edition (64-bit) - Microsoft SQL Server 2014

Server Explorer

AS

Server Objects

Database

book_dbs

bookflow_db

bookflow_dbs

joins

Tables

class

class_info

student_marks1

Views

Stored Procedures

Functions

Schema

class

class_info

student_marks1

joins

joins*

```
84 ON class.id=class_info.id;
85
86 SELECT *
87 FROM class
88 RIGHT JOIN class_info
89 ON class.id=class_info.id
90 WHERE class.id IS NULL;
91
92
93 CREATE TABLE first_table/
```

Result Grid

	id	name	id	address
▶	NULL	NULL	7	NOIDA
	NULL	NULL	8	PANIPAT

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SQL Server Enterprise Edition (64-bit) - Microsoft SQL Server 2008 R2

Server: <server>

AS

Database objects

- bank
- book_dbs
- bookflow_db
- bookflow_dbs
- joins
 - Tables
 - class
 - class_info
 - student_marks1
 - Views
 - Stored Procedures

joins x

```
93 CREATE TABLE first_table(  
94     id INT,  
95     name VARCHAR(30)  
96 );  
97  
98 CREATE TABLE second_table(  
99     id INT,  
100    name VARCHAR(30)  
101 );
```

18 12:17:01 CREATE TABLE first_table(id INT, name VARCHAR(30))

19 12:17:02 CREATE TABLE second_table(id INT, name VARCHAR(30))

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Edit View Query Database Server Tools Scripting Help

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er objects

bank

book_dbs

bookflow_db

bookflow_dbs

joins

Tables

class

class_info

student_marks1

Views

Stored Procedures

Functions

kleb

klef

stration Schemas

ation

object selected

joins*

106

107 • INSERT INTO second_table VALUES

108 (2, 'adam'),

109 (3, 'chester');

110

111 • SELECT * FROM first_table

112 UNION

113 SELECT * FROM second_table;

114

115 • SELECT name FROM first_table

Result Grid

Filter Rows:

	id	name
▶	1	abhi
	2	adam
	3	chester

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File Edit View Query Database Server Tools Scripting Help

SQL Server Enterprise Edition

Server Explorer

AS

Server Objects

Server

Database

Tables

class

class_info

student_marks1

Views

Stored Procedures

Functions

joins*

```
112 UNION
113 SELECT * FROM second_table;
114
115 • SELECT name FROM first_table
116 UNION
117 SELECT name FROM second_table;
118
119 • SELECT * FROM first_table
120 UNION ALL
121 SELECT * FROM second_table;
```

Result Grid

	name
▶	abhi
	adam
	chester

object selected

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File Edit View Query Database Server Tools Scripting Help

SQL Server Enterprise Edition

Server Explorer

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Server Objects

Server

Database

Tables

class

class_info

student_marks1

Views

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Functions

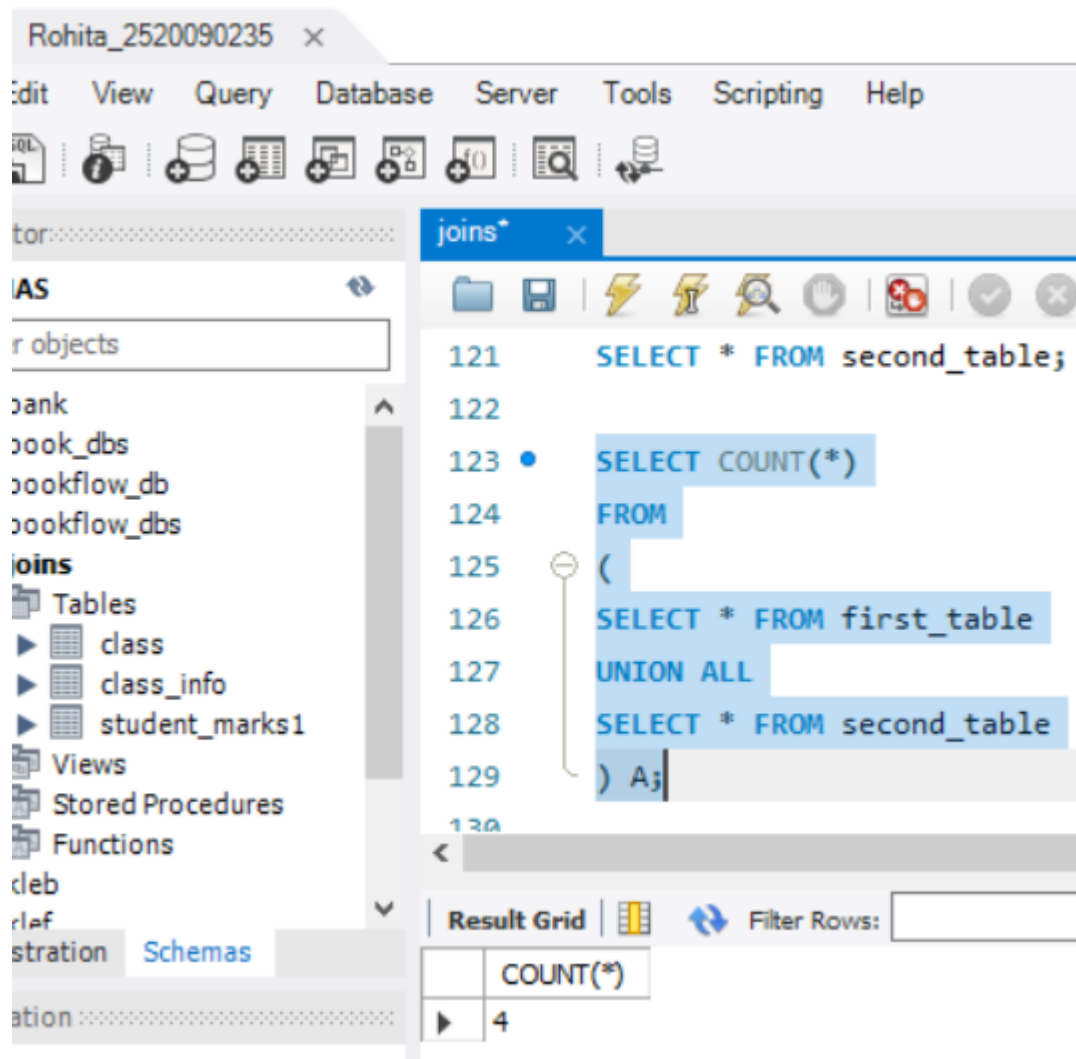
joins*

```
115 • SELECT name FROM first_table
116 UNION
117 SELECT name FROM second_table;
118
119 • SELECT * FROM first_table
120 UNION ALL
121 SELECT * FROM second_table;
122
123 • SELECT COUNT(*)
124 FROM
```

Result Grid

	id	name
▶	1	abhi
	2	adam
	2	adam
	3	chester

object selected



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edit View Query Database Server Tools Scripting Help

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objects

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book_dbs
bookflow_db
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oins
Tables
class
class_info
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Views
Stored Procedures
Functions
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ation

joins x

127 UNION ALL
128 SELECT * FROM second_table
129) A;
130
131 • SELECT * FROM first_table
132 INTERSECT
133 x SELECT * FROM second_table;
134
135 • SELECT name FROM first_table
136 INTERSECT
137 x SELECT name FROM second_table;

Result Grid Filter Rows:

	id	name
▶	2	adam

Rohita_2520090235 x

File Edit View Query Database Server Tools Scripting Help

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objects

ank

ook_dbs

ookflow_db

ookflow_dbs

oins

Tables

class

class_info

student_marks1

Views

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Functions

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lef

kill2

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tion

joins x

129) A;

130

131 • SELECT * FROM first_table

132 INTERSECT

133 ✖ SELECT * FROM second_table;

134

135 • SELECT name FROM first_table

136 INTERSECT

137 ✖ SELECT name FROM second_table;

138

139 • SELECT * FROM first_table

Result Grid

	name
▶	adam

Rohita_2520090235 x

Edit View Query Database Server Tools Scripting Help

SQL

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IAS

or objects

bank

book_dbs

bookflow_db

bookflow_dbs

joins

Tables

- class
- class_info
- student_marks1

Views

Stored Procedures

Functions

kleb

klef

skill2

stration Schemas

ation

joins x

135 • SELECT name FROM first_table

136 INTERSECT

137 ✖ SELECT name FROM second_table;

138

139 • SELECT * FROM first_table

140 ✖ EXCEPT

141 SELECT * FROM second_table;

142

143 • SELECT c.id,c.name,ci.address

144 FROM class c

145 INNER JOIN class_info ci

Result Grid

	id	name
▶	1	abhi

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Edit View Query Database Server Tools Scripting Help

SQL Server Enterprise Edition (64-bit) - Microsoft SQL Server 2014

Server Explorer

Server Objects

bank

book_dbs

bookflow_db

bookflow_dbs

joins

Tables

class

class_info

student_marks1

Views

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kleb

klef

skill2

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Schemas

ation

joins*

137 SELECT name FROM second_table;

138

139 SELECT * FROM first_table

140 EXCEPT

141 SELECT * FROM second_table;

142

143 SELECT c.id,c.name,ci.address

144 FROM class c

145 INNER JOIN class_info ci

146 ON c.id=ci.id;

147

Result Grid

Filter Rows:

Export

	id	name	address
▶	1	abhi	DELHI
	1	abhi	DELHI
	1	abhi	DELHI
	1	abhi	delhi
	1	abhi	delhi
	2	adam	MUMBAI
	2	adam	MUMBAI
	2	adam	MUMBAI
	2	adam	mumbai
	2	adam	Mumbai
	1	abhi	DELHI

object selected

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Edit View Query Database Server Tools Scripting Help

SQL Server Enterprise Edition (64-bit) - Microsoft SQL Server 2014

Server Explorer

AS

Server objects

Bank

book_dbs

bookflow_db

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joins

Tables

class

class_info

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cleb

clef

skill2

stration

Schemas

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joins*

148 SELECT c.id,

149 c.name,

150 CASE

151 WHEN ci.address IS NULL

152 THEN 'Address Missing'

153 ELSE 'Address Available'

154 END AS Status

155 FROM class c

156 LEFT JOIN class_info ci

157 ON c.id=ci.id;

Result Grid

Filter Rows: Exp

	id	name	Status
▶	1	abhi	Address Available
	1	abhi	Address Available
	1	abhi	Address Available
	1	abhi	Address Available
	1	abhi	Address Available
	2	adam	Address Available
	2	adam	Address Available
	2	adam	Address Available
	2	adam	Address Available
	2	adam	Address Available
	4	alex	Address Missing

object selected